DevOps for Cyber-Physical Systems

AGENTIC MULTI-UAV COORDINATION USING LLMS

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PROBLEM & MOTIVATION

Motivation

- Large Language Models (LLMs) are emerging as high-level reasoning engines for robotics
- Multi-UAV systems require autonomous coordination and task allocation
- Failure modes in LLM-driven multi-agent systems are still not well understood

Problem Statement

- How can multiple LLM-driven UAVs coordinate robustly under communication failures, conflicting instructions, and adversarial behavior?

TECHNOLOGIES & SYSTEM COMPONENTS

LLM (Mission Planning) → Converts natural language missions into structured UAV tasks.

Python (Coordinator Agent) → Implements the central coordinator that interacts with the LLM and manages task assignment.

2 **ROS2 (Communication Layer)** → Handles message passing between the coordinator and UAV agents.

PX4 SITL (Drone Controller) → Simulated drone flight controller that executes waypoint commands.

Gazebo → Simulates the multi-UAV environment and drone interactions.

Docker → Containerizes the system to ensure reproducibility and consistent deployment.
Docker containers run → ROS2, PX4 SITL, Gazebo & UAV agents

WORKFLOW

Conceptual	Technical
High-level mission intent	Natural language parsing & constraint extraction
Multi-agent task allocation	Centralized task allocation using LLM coordinator (MVP)
Centralized task dispatching	ROS2 messaging between coordinator and UAV agents
Collaborative mission execution	Waypoint generation → ROS2 → PX4 SITL → Gazebo simulation (3 UAVs)

We keep the LLM-based coordination only (no VLM/camera perception) to keep scope feasible for a 2-person team

EXPERIMENTS

Functional Testbed (the system we build)

- 3 UAVs simulated using PX4 SITL and Gazebo
- LLM coordinator assigns tasks to UAV agents
- UAVs execute tasks via ROS2 communication

Test Scenarios (our experiments)

- Formation flight (Leader–Follower): UAV1 follows waypoints; UAV2 and UAV3 track fixed offsets relative to UAV1.

Metrics & Evaluation (measurements of success)

- Formation error (avg): mean distance from each follower to its desired offset position
- Formation error (max): worst-case deviation during flight
- Mission completion time: time for the leader to finish the waypoint route
- Success rate: % runs where formation error stays below a threshold (e.g., < 2m)

THANK YOU

Agentic Multi-UAV Coordination using LLMs